































BLADE ADAPTER V0.1 / 5V addressable RGB / local TIM3 DMA control / NOT ROUTED

J701 C32713227 horizontal male. Pins: 1=SDA, 2=GND/DET, 3=GND, 4=SCL, 5=5V_ACC, 6=3V3. Mate J7 pins 5,6,3,4,1,2 respectively. Power OFF before mating; shell must key and support connector.

PB6 pin20 = I2C1_SCL AF6; PB7 pin1 = I2C1_SDA AF6; other GPIOs bonded to these pins must remain analog inputs. Slave address 0x32.

PA6 pin13 = TIM3_CH1 AF1 LED waveform; PA4 pin11 = LED_ENABLE. PA13/PA14 = SWD. Bare test pads TP701-TP705: SWDIO, SWCLK, NRST, 3V3, GND.

Boot/reset/communication timeout: LED_ENABLE=0. Require valid identity/configuration and current budget before arm. STM32 96-bit UID is identity, not authentication.

LED type/count currently UNCONFIGURED. Firmware must refuse arm until a compatible timing profile and count are configured. Host budget <=400mA at 5V; logic <=50mA at 3.3V.

TPS22919 gates LED rail; SN74LVC1T45 VCC isolation avoids driving DIN while 5V_LED is off. J702: 1=5V_LED,2=DIN,3=GND. No reverse-power input.

Prototype for keyed single-accessory mating. Power off before insertion/removal; hot-plug sequencing/ESD and actual LED load require bench validation.